Precision and Recall

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Precision and Recall before opening notes. Show any later correction separately.

Try the **recall** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Precision and Recall trace and record the first step you can explain without help.

Use W2: Calculate both ratios for **Precision recall** and **denominator**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Precision and Recall explanation against the supplied facts before discussing it.

Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Precision and Recall target to a trace and a justified conclusion.

TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

WORKBOOK · 6 OF 21

Solve the given case

Solve the supplied Precision and Recall case and show the deciding evidence.

Positive is spotted: TP=8, FP=2, FN=4.

Compute **precision** and **recall**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or **miss**ing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Precision and Recall adds to Responsible Plant Classifier and keep the evidence.

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Identify the groups

Identify the groups. Apply the supplied Precision and Recall facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Precision and Recall", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked matrix, give predicted-positive and reference-positive totals.

My reasoning and evidence

W2 Calculate both ratios

> Calculate both ratios. Apply the supplied Precision and Recall facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Precision and Recall", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Calculate **precision**, **recall** and accuracy for the worked matrix.

> **My reasoning and evidence**

W3 Handle no flags

> Handle no flags. Apply the supplied Precision and Recall facts and keep your reasoning.

Explain **precision** and **recall** for [[16, 0],[4, 0]].

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Precision and Recall facts and keep your reasoning.

Rules supplied for this independent case

Matrix convention for this exercise: rows are reference classes [clear, blocked]; columns are predicted classes [clear, blocked]. Blocked is the positive class, so the layout is [[TN, FP], [FN, TP]].

Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit a claim

> Limit a claim. Apply the supplied Precision and Recall facts and keep your reasoning.

A team reports **recall** 100 percent but hides **precision** and counts. What is **miss**ing?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

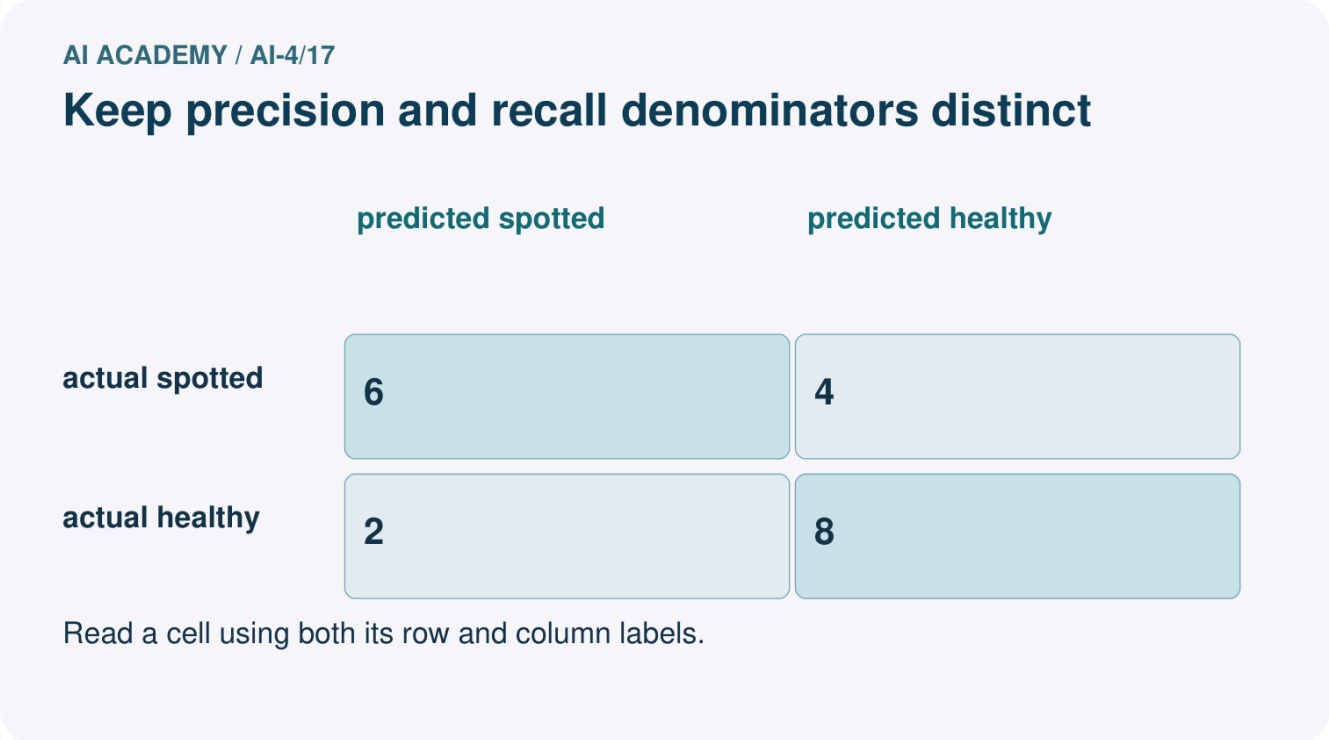
Pictorial case practice: Precision and Recall

> Put precision and recall denominators on the picture instead of memorising names. Explain the labelled step before checking the changed condition.

Put **precision** and **recall denominator**s on the picture instead of memorising names.

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/17

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Predicted positives

Eight leaves were flagged: precision=6/(6+2)=75%.

↓

2. Actual positives

Ten leaves were spotted: recall=6/(6+4)=60%.

↓

3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Put **precision** and **recall denominator**s on the picture instead of memorising names.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

1. Predicted positives

2. Actual positives

3. Keep goals separate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep goals separate **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Read the picture in order.

Precision asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Repair a missing building block

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

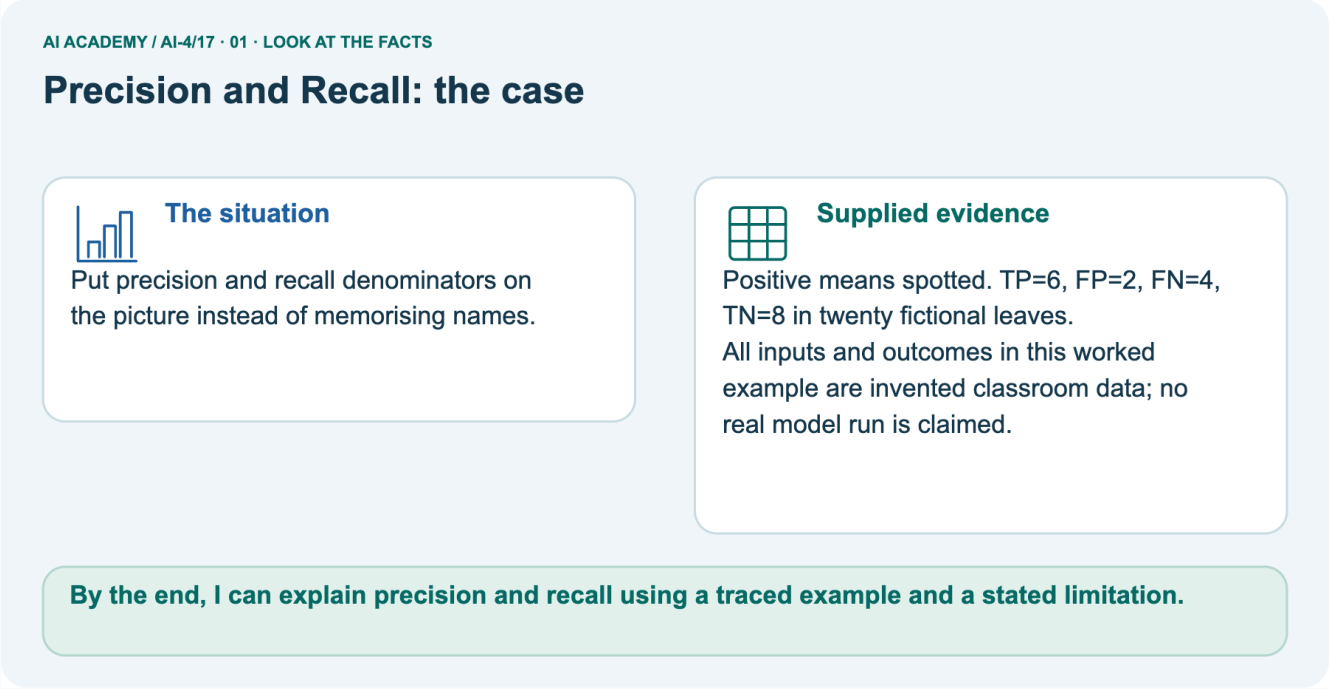
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Precision and Recall picture from its labels, then record a relevant limit.

01. Precision and Recall: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

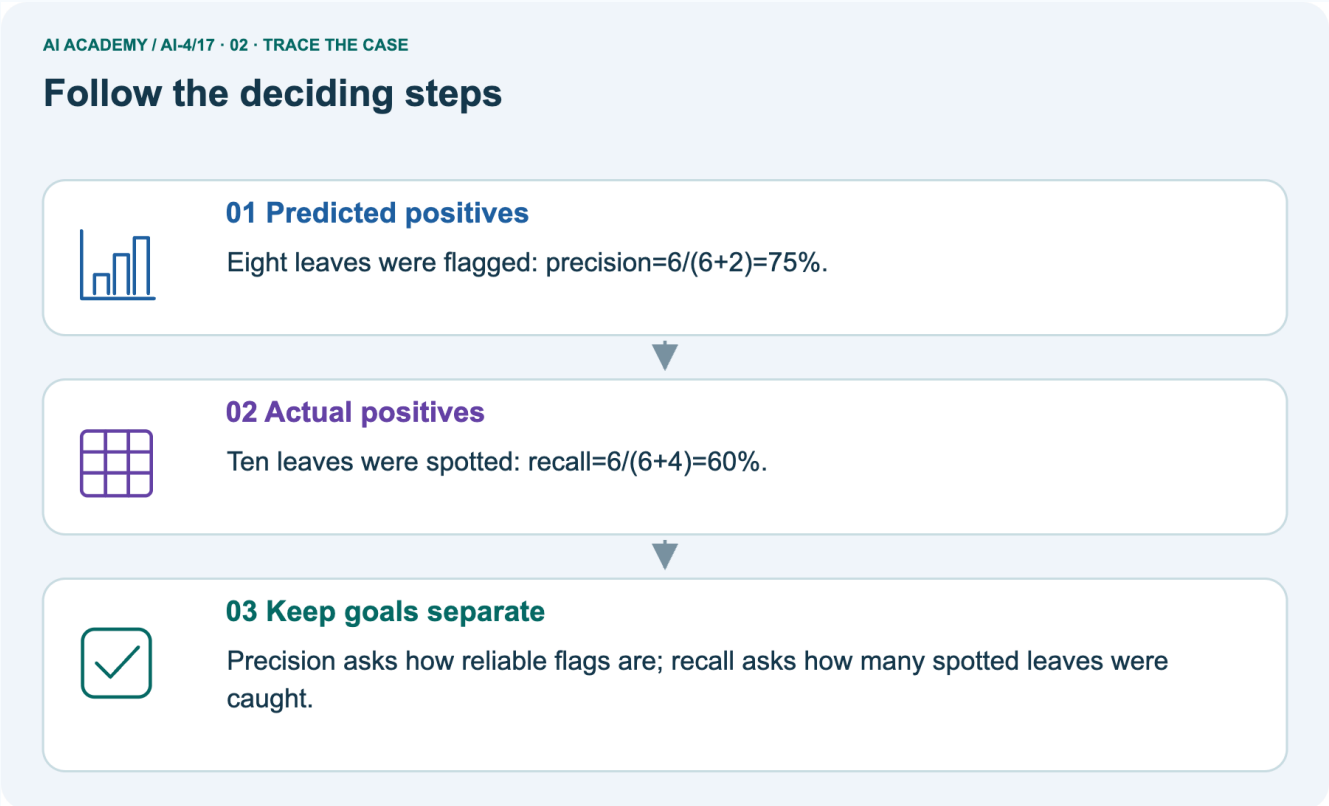
Put **precision** and **recall denominator**s on the picture instead of memorising names. Supplied facts: Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



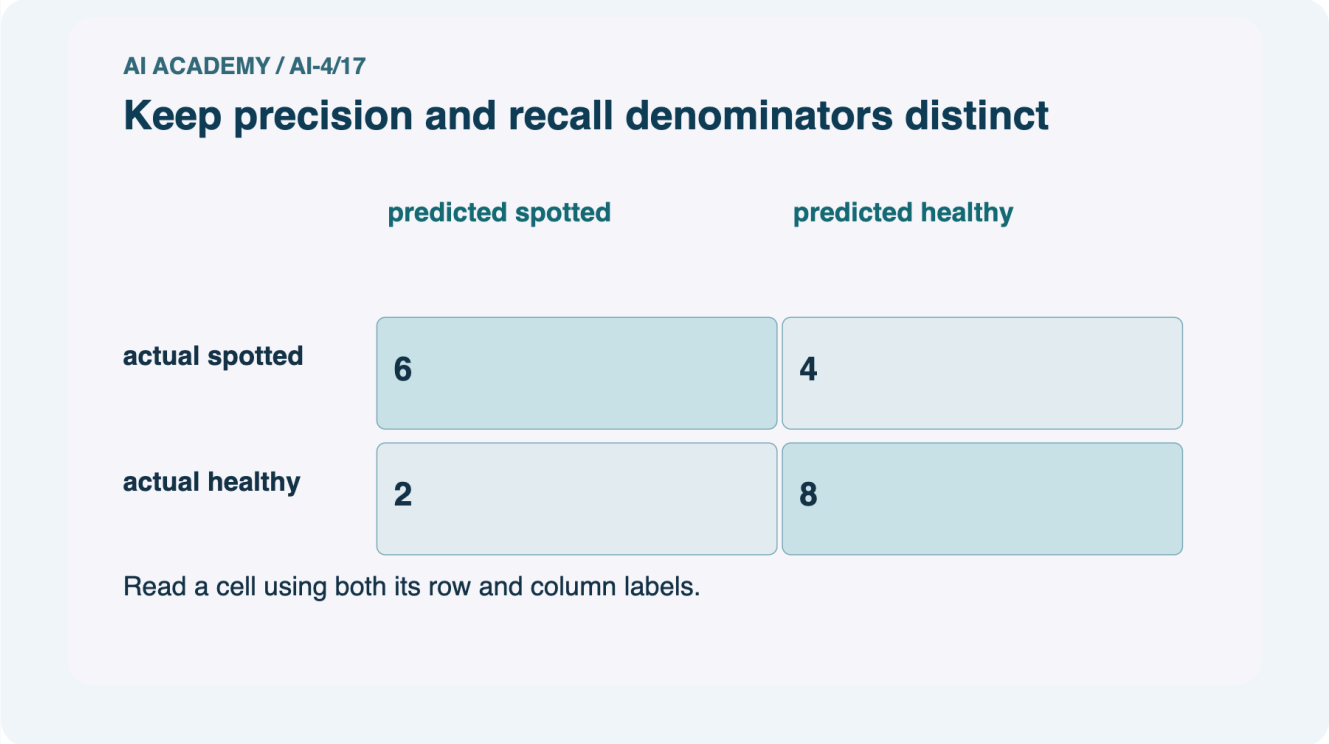
Follow the labelled picture with the matching explanation. Open the vector diagram

Predicted positives: Eight leaves were flagged: **precision**=6/(6+2)=75%. → Actual positives: Ten leaves were spotted: **recall**=6/(6+4)=**60%**. → Keep goals separate: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep precision and recall denominators distinct



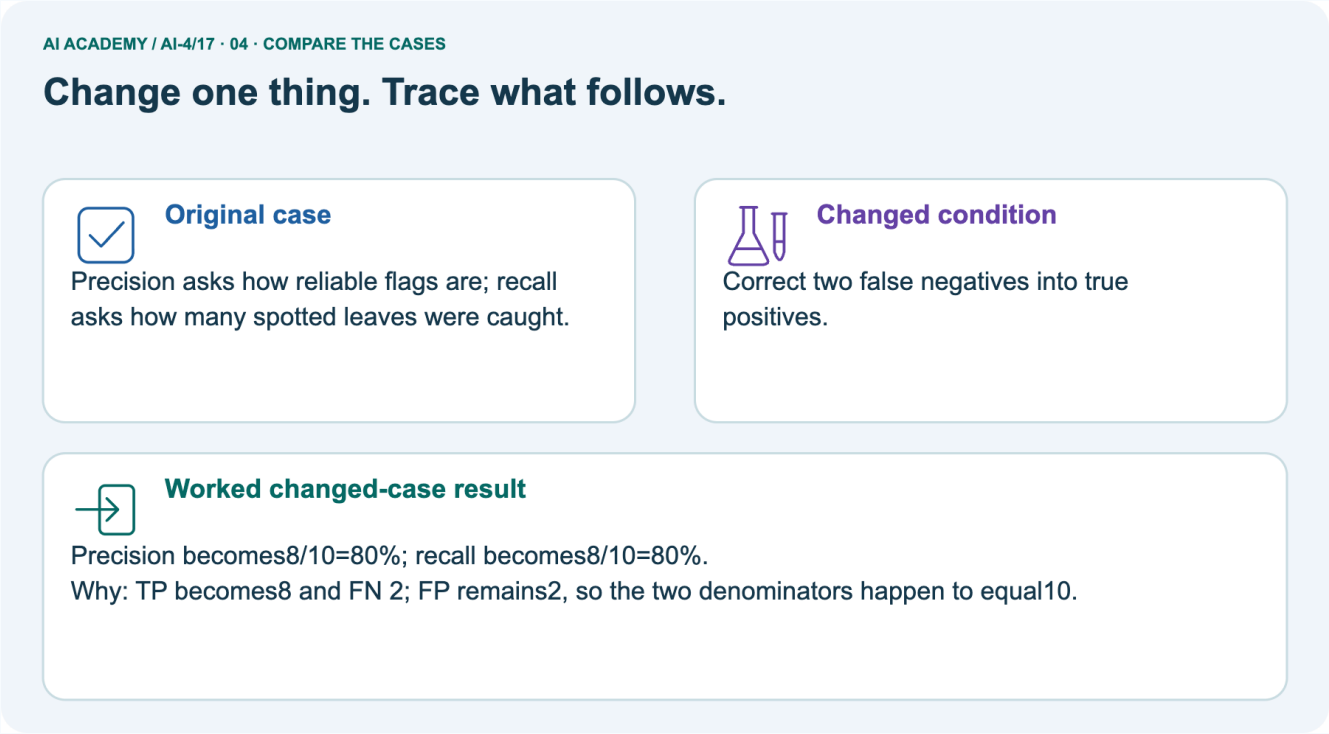
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **precision** and **recall denominator**s distinct. actual spotted: predicted spotted = 6, predicted healthy = 4. actual healthy: predicted spotted = 2, predicted healthy = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

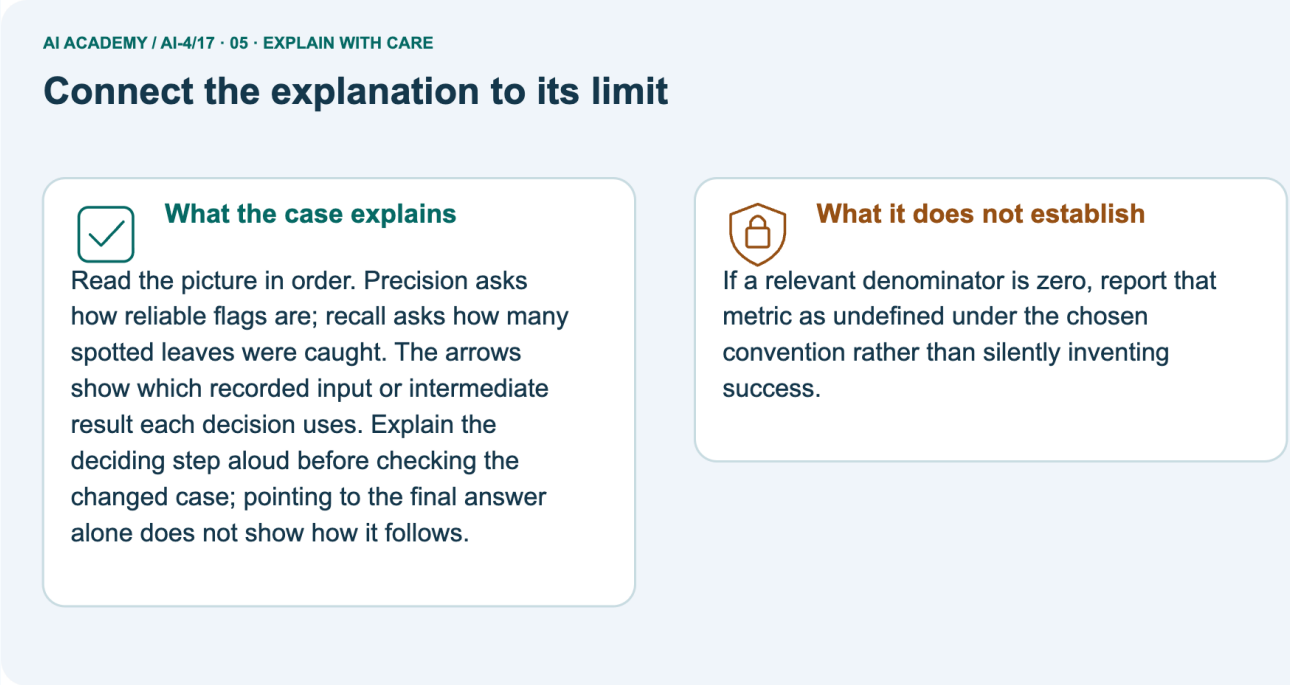
Original case: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Changed condition: Correct two false negatives into true positives.

Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

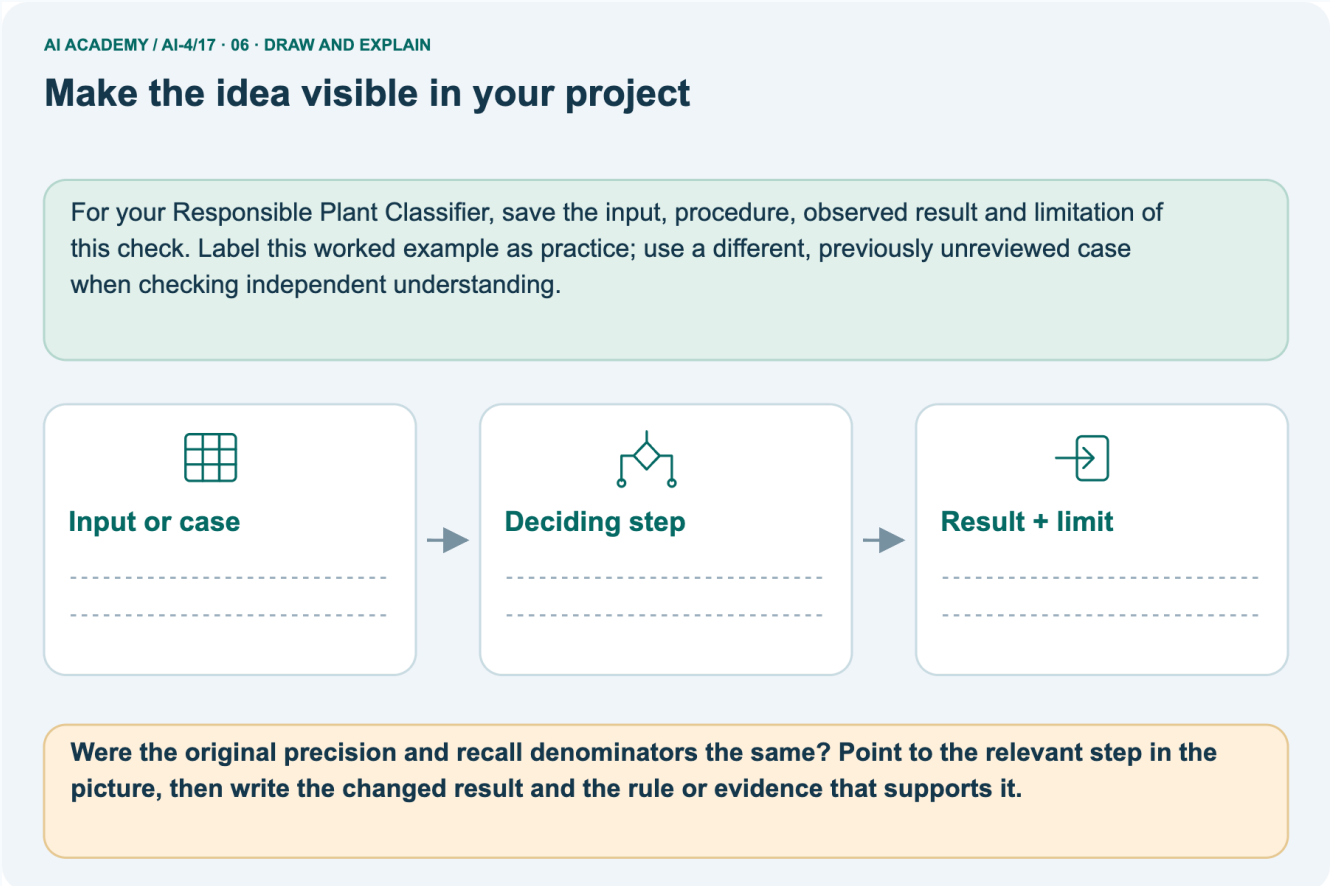
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.